



AI Agent with Voiceflow for Operating Manuals

Optimization of business processes through Artificial Intelligence

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Impact

@puli





MANTA Group operates in the aerospace, automotive and industrial sectors, with consolidated experience in the assembly of complex parts and sub-assemblies, in the production of composite components, in precision machining, in NDT testing and in surface treatments.

In the civil aerospace sector, the company contributes to important international programs, including: Boeing 767, AW139, AW169, AW189 and AW609.



**MICHELE
FRISOLI**

The problem: operational inefficiency

Complex and Difficult Manuals

The current situation sees operators forced to consult paper or PDF manuals of over 200 pages to find urgent procedures.



Wasted time: Employees spend up to 20% of their time searching for information.



Risk of Bias: Difficulty in accessing leads to relying on memory, increasing operational risk.



Objectives and Strategy



Accessibility

Reduce search time from minutes to seconds, making company knowledge immediately available.



AI solution

A conversational agent powered by Voiceflow that understands natural language and responds like an expert.



RAG technology

Retrieval Augmented Generation

AI reads real PDF manuals, integrates customer-specific requests, and generates precise responses by quoting:

Type of manual

Paragraph

Reference pages



Secondary Objectives

Dual-language translation of
documentation

Integration with company calendar to report maintenance,
activities, revisions, etc.

Advanced Data Protection

Upgrade al piano Voiceflow
Enterprise



Technical Architecture

1

User Query

"How do I reset the X
valve?"
(Voice or text input)

2

Voiceflow NLU

Intent analysis and
knowledge base activation.

3

Retrieval (RAG)

Extract relevant paragraphs
from indexed PDF
documents.

4

Generated Response

LLM summarizes the
answer and provides the
link to the page.

User Experience: Demo



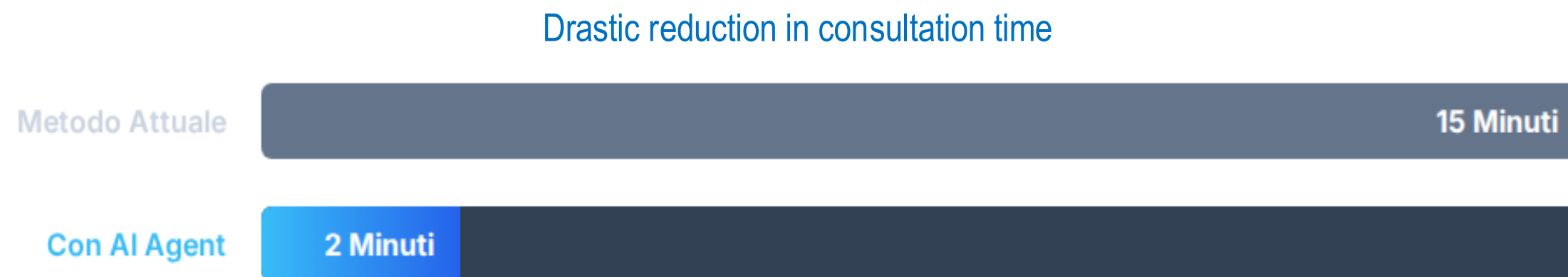
The Assistant always in your pocket

The use scenario involves a maintenance technician in the field. Instead of going back to the office to look up the manual, send a voice message. The agent, after analyzing the customer's specific requests, responds instantly with the step-by-step procedure.

Advantage: On-the-job training and real-time problem solving.



Impact and Conclusions



The project demonstrates how AI can transform static documentation into a dynamic asset, improving efficiency and security and analyzing specific client requests before indicating a solution.

SOLUTION: CONVERSATIONAL AI AGENT

The AI agent supports the operator in quickly consulting the operating manuals, providing immediate and contextualized answers. Based on the Voiceflow platform;

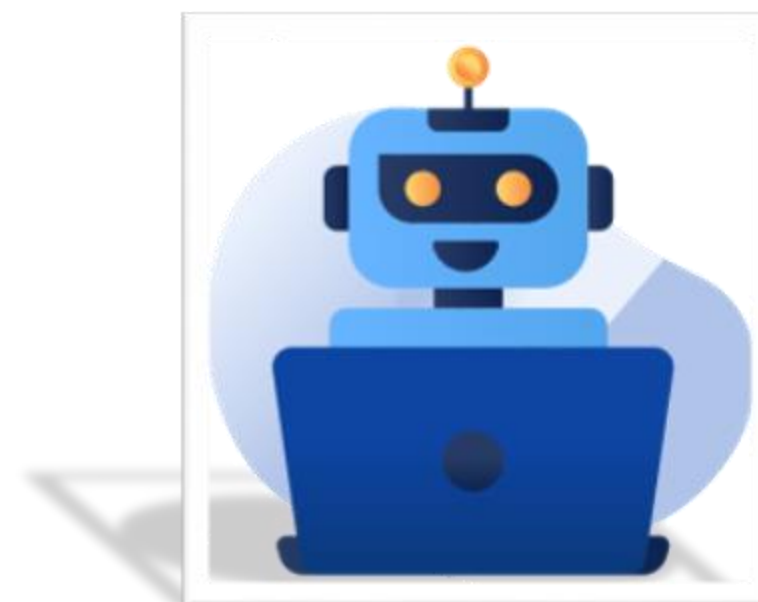
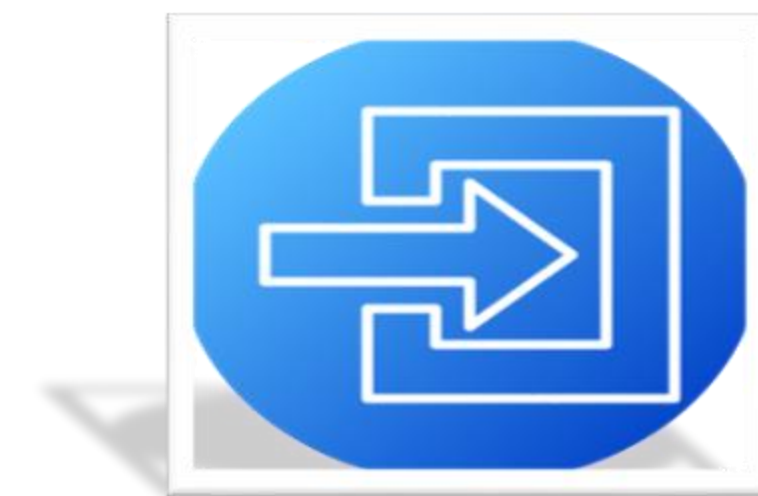
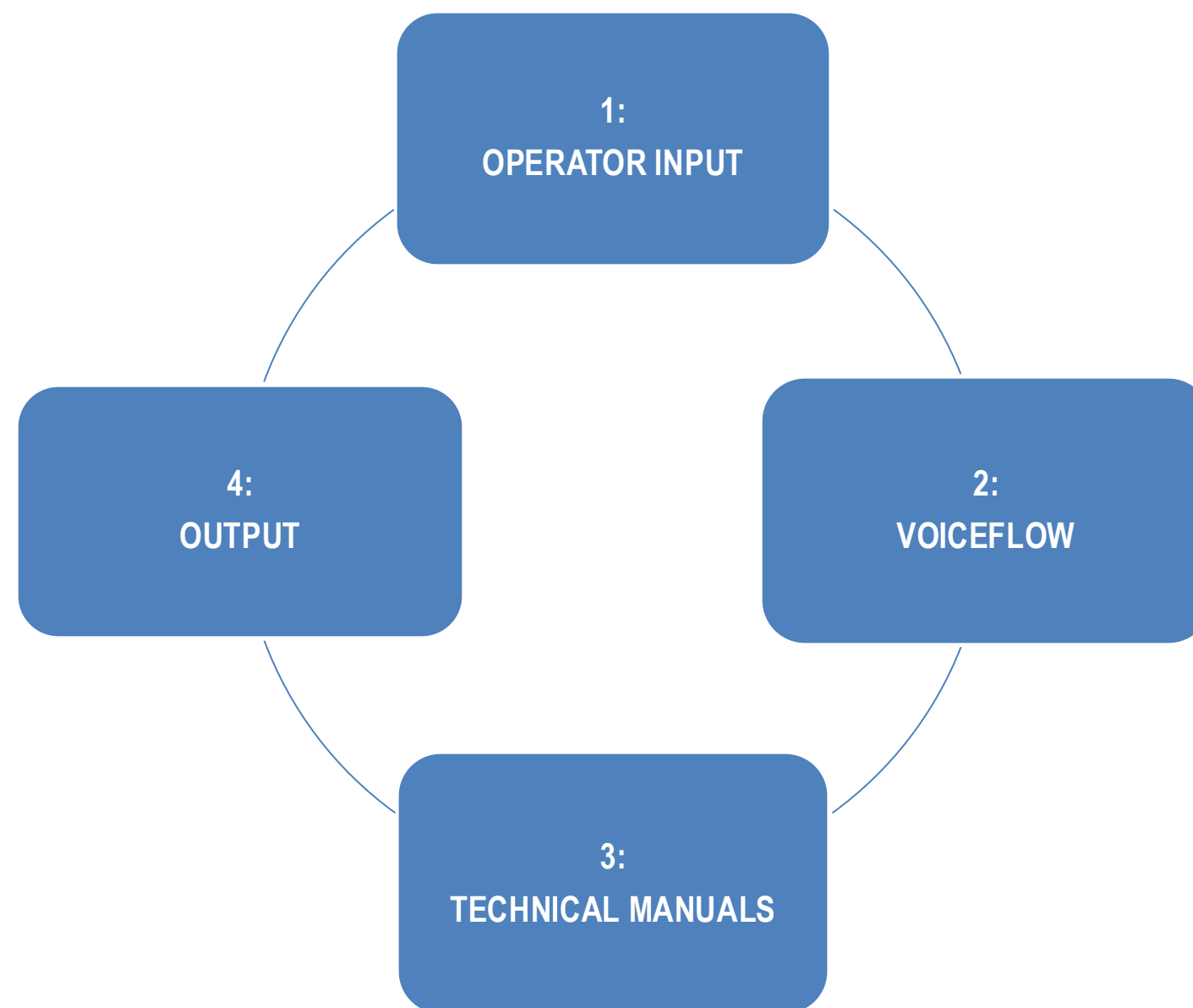
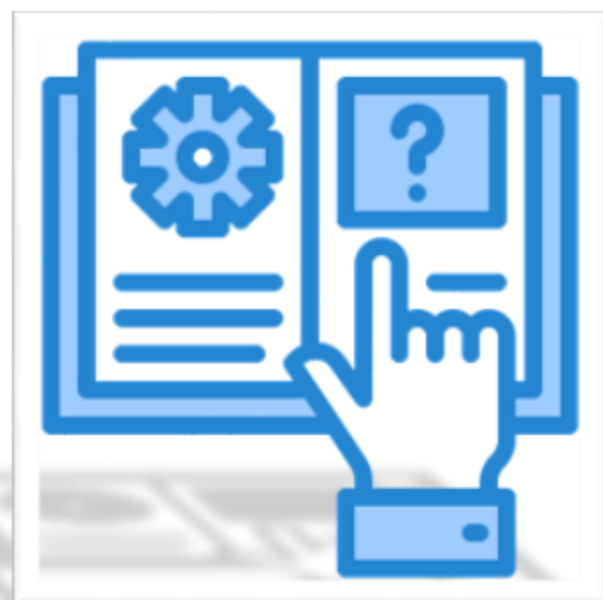
Allows natural voice interaction with the operator;

Accesses internal operating manuals to provide accurate information;

Report each answer citing manual, chapter and reference paragraph.



System architecture



Feature: Contextualized Responses

1. Voice responses based on verified content in operating manuals.
2. Precise references to: "Manufacture of Parts".
3. Significant reduction in search time by the operator.

AI Responses

According to the manual "Manufacture of parts", Chap.3-Par. 3.2, The maximum orientation tolerance for plies in flat parts is:

Fabric: $\pm 2^\circ$;

Tape: $\pm 1^\circ$.

Reference:

Table "Tolerances Orientation Lamination", page 47.

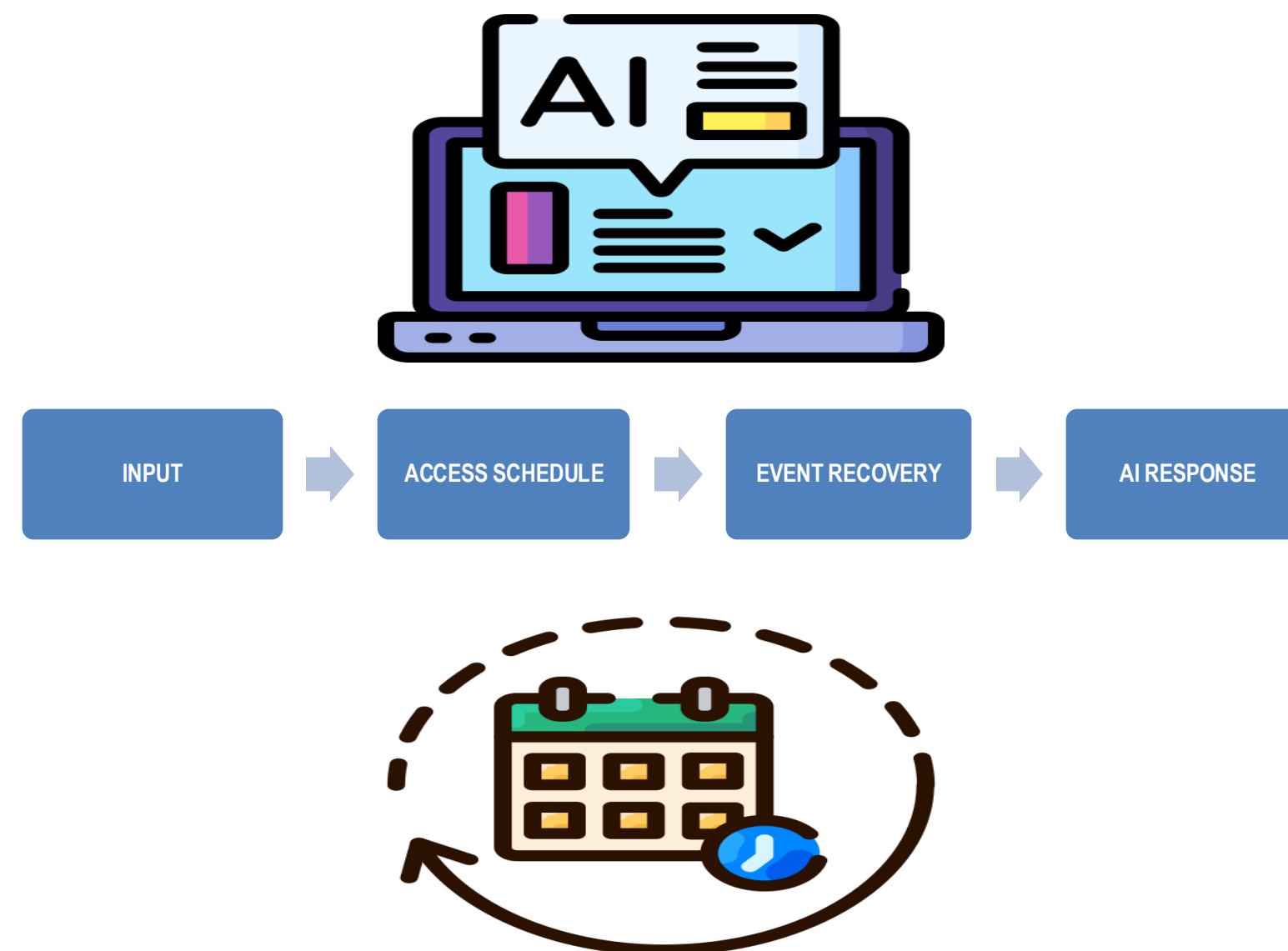
TRANSLATION INTO TWO LANGUAGES

- Automatic translation of technical content into the language required by the operator.
- Immediate support for companies with international staff.
- Reduction of interpretative errors in the operational phases.



Integration with the timetable

The AI agent can access and consult the company schedule to provide information on maintenance, planned activities, overhauls and operational deadlines, improving the time management of the operating staff.





Integration with the Business Management System

The integration of the AI agent with the management system would make it possible to automatically link the customer's request to the internal technical documentation, suggesting to the operator the most suitable procedure for the specific case.

Operator Input

"Customer X requests info on component Y"

"What is the status of job Z?"

"What is the procedure associated with the detected defect?"

AI + Management Processing

Search in the management system → order → related activities → documentation

Query to the management system → extraction progress

Defect matching → procedure → technical paragraph

Output to Employee

Correct Manual + Chapter + Applicable Instruction

Updated status + missing tasks

Correct statement + page + warning



Benefits for the Operations Department

Operational Efficiency	Quality of Information	Value for Staff
Reduction of the time spent consulting manuals.	Reduction of operational errors.	Facilitation of training of new operators.
Instant access to critical information.	Consistency of the procedures applied.	Constant support in daily activities.



Introduction to the Demo

This demo shows the AI Agent in action: interpret MANTA procedures, standards and technical documentation with total reliability. It reduces search times, increases the quality of answers and supports operational decisions. A real-world example of how AI can become part of the daily workflow.





Demo - Example 1

QUESTION:

Can you briefly list the materials in contact with the part in the composite parts manufacturing manual?

 **TECHNICAL ANSWER:** Materials in contact with the part, as outlined in the composite parts manufacturing manual, include: Release Film: WW3400 R, Manufacturer: Firtech Liquid Release Agent: Zicote TRE 45 ECO Non-Silicone Release Agent, Manufacturer: Zicote Italia Disposable Composite Protective Gloves (Nitrile, Powder-Free) These materials are essential for ensuring the proper workmanship and quality of composite parts.

 **REFERENCES:** Document: POL-ENG-17 - MANUFACTURE OF COMPOSITE PARTS Paragraphs: 7.3.1 MATERIALS IN CONTACT WITH THE PART



Demo - Example 2

QUESTION:

What instruments are expiring in the next 10 days?

TECHNICAL ANSWER:

The following instruments are due to expire in the next 10 days: TOO-0602: Digital Vernier Caliper 0-150/0.01 mm, Expiration: 5/11/2025 HE21-001905: Digital Thermo-Hygrograph, Expiration: 01/12/2025 These instruments should be checked and, if necessary, recalibrated or replaced to ensure their functionality and compliance.



Challenges Faced

Data integration

Building the AI Agent required connecting manuals and procedures in heterogeneous formats and importing documents into Voiceflow. A work of standardization and adaptation that has guaranteed consistency and accessibility of information.

Security and privacy

During development, critical topics such as access control and protection of sensitive data were addressed. The final solution integrates advanced security policies and clear processes to ensure that technical information remains protected.



Challenges Faced

Building the foundation of intelligence

An effective AI Agent is not only born from the integration of documents, but from a carefully designed logical structure: clear paths, well-defined intents and rules that guide interaction.

Targeted language training

To make the interaction natural and useful, the agent was trained to recognize free commands, synonyms and the typical formulations of MANTA technicians. A linguistic work that allows AI to really understand the operational context and respond accurately.





Future Developments: From Pilot to Digital Ecosystem

We explored the secure architecture and current features. The next technical priority is to scale the Agent from the Proof of Concept phase to the secure production environment.

Key Feature Expansion



Total Document Integration

Upgrade to VoiceFlow Enterprise for compliance and legal validity of responses.



Integration with the Management System

Connection to ERP/MES systems for automation, controls and proactive logistic warnings.



Advanced AI and Continuous Learning

Hands-free speech recognition on the shop floor and improve accuracy through interactions.





Strategic Extension: Applying the RAG Model to Other Processes

The modular and secure architecture of our Co-Pilot, based on the RAG methodology, can be extended to other document-intensive business domains.



HR Training and Onboarding

Immediate support on company policies and security procedures.



CV Screening & IT Support

CV classification for skills and IT support based on internal manuals.



Connections with IoT systems

Real-time data cross-referencing with regulatory limits for predictive diagnosis.

Measurable Benefits and Return on Investment (ROI)

Improved operational efficiency

Estimated 30-40% reduction in analysis and reporting time.

Cost and Error Reduction

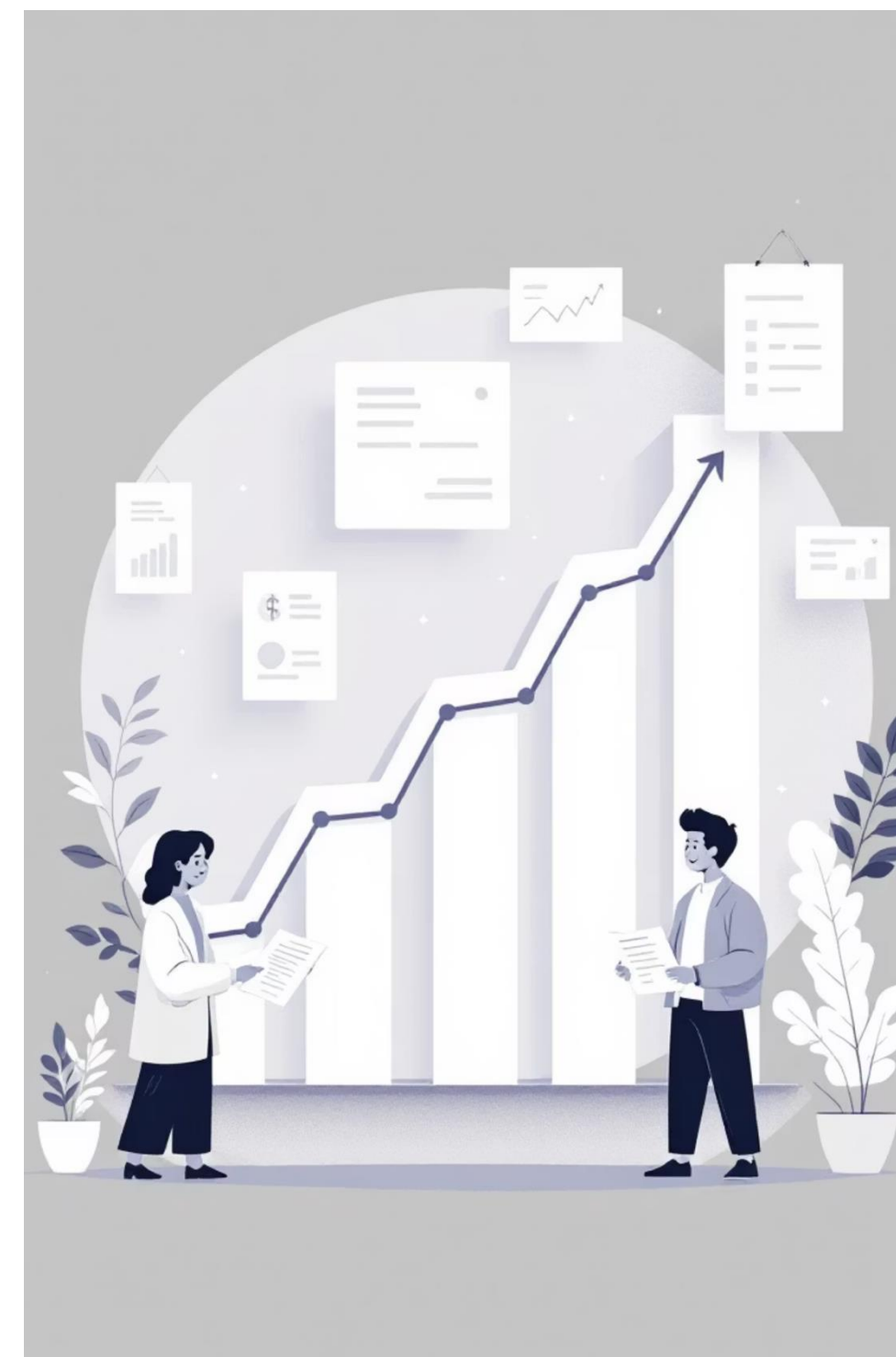
Prevention of Non-Conformities and goal zero transcription errors.

Increased Personal Satisfaction

Quick access to information reduces stress and makes procedures easier.

Increased Audit Readiness

Automatic traceability of responses ensures document compliance.





Implementation Roadmap: Short-Medium Term Milestones

Phase 1: Pilot Implementation

Enterprise environment upgrade, Vector Store integration, Composite department pilot launch, feedback collection.

1

2

Phase 2: Integration and Scalability

Implementation of additional features, integration with ERP/MES systems, expansion to other departments.

